

0625 • PAPER 4

TOPIC

06

Space Physics

Questions only • Reordered into focused subtopic sections

27

QUESTIONS

—

TOTAL MARKS

7

SECTIONS



USE THE TRACKER • ATTEMPT WITHOUT NOTES • MARK • REVIEW ERRORS

TOPIC CONTENTS

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6.1 Planets and the Solar System

#	SOURCE	MARKS	DONE	MARK	REVIEW
01	Oct/Nov 2025 Paper 41 Question 2	-	☐	☐	☐

- 2 Table 2.1 contains information about the planet Mars.

Table 2.1

mass	$6.4 \times 10^{23} \text{ kg}$
gravitational field strength at surface	3.7 N/kg
average density	3900 kg/m^3

- (a) Define gravitational field strength.

.....
 [1]

- (b) (i) An object has a weight of 42 N at the surface of the Earth.

Calculate the weight of the object at the surface of Mars.

weight = [2]

- (ii) Calculate the volume of Mars.

volume = [2]

- (c) Fig. 2.1 shows a space buggy that is tested on Earth. The buggy is travelling at a constant speed in a straight line. The driving force on the buggy is 30 N.

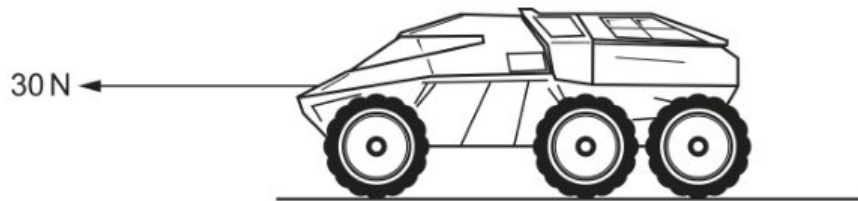


Fig. 2.1

- (i) Draw and label **one** arrow on Fig. 2.1 to show the size and direction of the resistive forces on the buggy. [2]
- (ii) Air resistance on Mars is less than air resistance on Earth. The same driving force, 30 N, is exerted on the buggy on Mars.

1. State the effect this has on the resultant force on the buggy on Mars.

.....

2. State the relationship between resistive forces, driving force and resultant force.

.....

[1]

[Total: 8]

6.2 Moon phases and orbits

#	SOURCE	MARKS	DONE	MARK	REVIEW
01	Feb/March 2024 Paper 42 Question 10	-	☐	☐	☐

- 10 (a) Fig. 10.1 represents different positions A–H of the Moon as it rotates around the Earth.

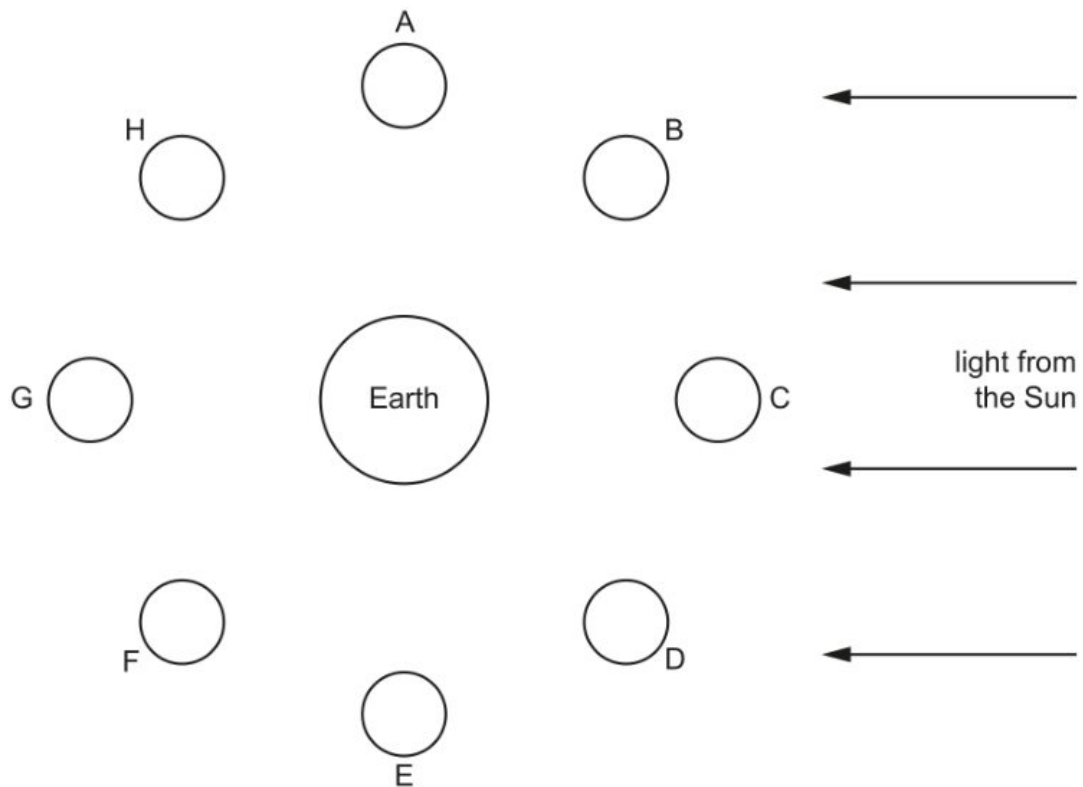


Fig. 10.1

- (i) State a position of the Moon where an observer on Earth sees:

1. there is a quarter Moon 

2. there is a full Moon 

[2]

- (ii) State the approximate time taken for the Moon to orbit the Earth.

time = [1]

(b) The average distance of the Earth from the Sun is 1.5×10^8 km.

(i) Calculate the average orbital speed of the Earth in km/h.

average orbital speed = km/h [3]

(ii) The speed of light in a vacuum is 3.0×10^8 m/s. Calculate the time taken for light from the Sun to reach the Earth.

time = [2]

[Total: 8]

6.3 Comets and orbits

#	SOURCE	MARKS	DONE	MARK	REVIEW
01	Oct/Nov 2025 Paper 43 Question 10	-	☐	☐	☐

- 10 (a) Sketch a diagram to show the orbital path of a comet around the Sun.

Mark the position of the Sun on your diagram using an S.

[2]

- (b) Draw a cross on your diagram to show where the comet is travelling with the slowest speed. Label the cross L. [1]

- (c) Explain why the speed of a comet changes as it travels around its orbit.

Use the conservation of energy in your explanation.

.....

.....

.....

..... [3]

[Total: 6]

6.4 Solar System and accretion model

#	SOURCE	MARKS	DONE	MARK	REVIEW
01	Feb/March 2023 Paper 42 Question 10	-	☐	☐	☐
02	May/June 2023 Paper 41 Question 10	-	☐	☐	☐
03	May/June 2023 Paper 42 Question 10	-	☐	☐	☐
04	May/June 2023 Paper 43 Question 9	-	☐	☐	☐
05	Oct/Nov 2023 Paper 42 Question 9	-	☐	☐	☐
06	May/June 2024 Paper 42 Question 10	-	☐	☐	☐
07	Oct/Nov 2024 Paper 41 Question 9	-	☐	☐	☐
08	Oct/Nov 2024 Paper 42 Question 11	-	☐	☐	☐
09	Feb/March 2025 Paper 42 Question 10	-	☐	☐	☐
10	May/June 2025 Paper 41 Question 10	-	☐	☐	☐
11	May/June 2025 Paper 42 Question 10	-	☐	☐	☐
12	Oct/Nov 2025 Paper 42 Question 10	-	☐	☐	☐
13	Oct/Nov 2025 Paper 43 Question 9	-	☐	☐	☐

- 10 (a)** The time taken for Mars to orbit the Sun is 690 Earth days. The average orbital radius of Mars is 2.28×10^8 km. An Earth day is 24 h.

Calculate the average orbital speed of Mars in km/s.

average speed = [3]

- (b)** State the shape of the orbits of the planets.

..... [1]

- (c)** Light from a distant galaxy is redshifted.

- (i)** Explain what is meant by redshift.

.....
..... [2]

- (ii)** State the quantity that the redshift of a galaxy is used to calculate.

..... [1]

[Total: 7]

- 10 Pluto is a dwarf planet. Fig. 10.1 shows the direction of motion of Pluto as it follows its elliptical orbit around the Sun.

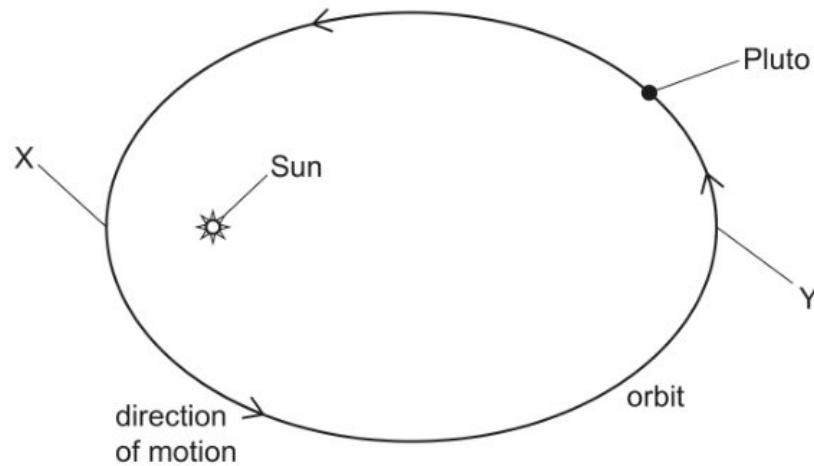


Fig. 10.1 (not to scale)

- (a) Point X is the point in the orbit closest to the Sun and point Y is the point furthest away.

The orbital speed of Pluto varies as it orbits the Sun.

- (i) Describe how the speed of Pluto varies as it moves from X to Y and then back to X.

.....
..... [1]

- (ii) Explain, in terms of energy transfers, why the speed of Pluto varies in this way.

.....
.....
.....
..... [3]

(b) The average temperature on the surface of Pluto is 43 K.

(i) Convert this temperature to a value in degrees Celsius ($^{\circ}\text{C}$).

temperature = $^{\circ}\text{C}$ [1]

(ii) Pluto has a white surface, as shown in Fig. 10.2. As Pluto rotates, the white surface alternately faces towards and away from the Sun.

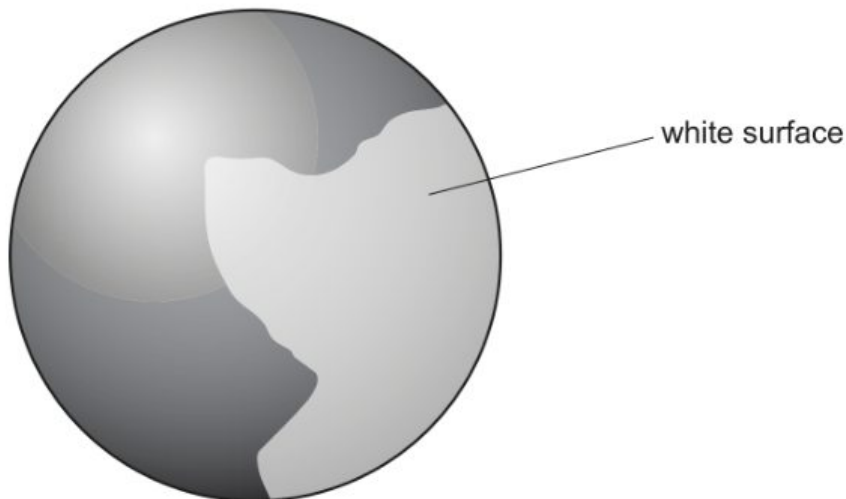


Fig. 10.2

Explain how this affects the temperature of Pluto as it rotates on its own axis.

.....
.....
.....
..... [2]

[Total: 7]

- 10 (a) State the equation that defines the average orbital speed v of a planet. State the meaning of any symbols you use.

.....
 [2]

- (b) Suggest why countries that are a significant distance from the Equator experience significant temperature variation throughout the year.

.....

 [1]

- (c) Fill in the gaps in the paragraph about a star much more massive than the Sun.

The stage that follows the stable state in the life cycle of the star is the

..... stage.

It then explodes as a supernova to form a, this leaves behind a

..... or a

[4]

- (d) A galaxy is moving away from the Earth with a speed of 33 000 km/s.
 The value of the Hubble constant is 2.2×10^{-18} per second.

Calculate the distance from the galaxy to the Earth. Give your answer in light-years.

distance = light-years [2]

[Total: 9]

- 9 Fig. 9.1 shows the Sun as the central dot and the planets Saturn, Jupiter and Earth labelled S_0 , J_0 and E_0 . The planets orbit the Sun anticlockwise. From the Earth's orbit, the planets appear aligned.

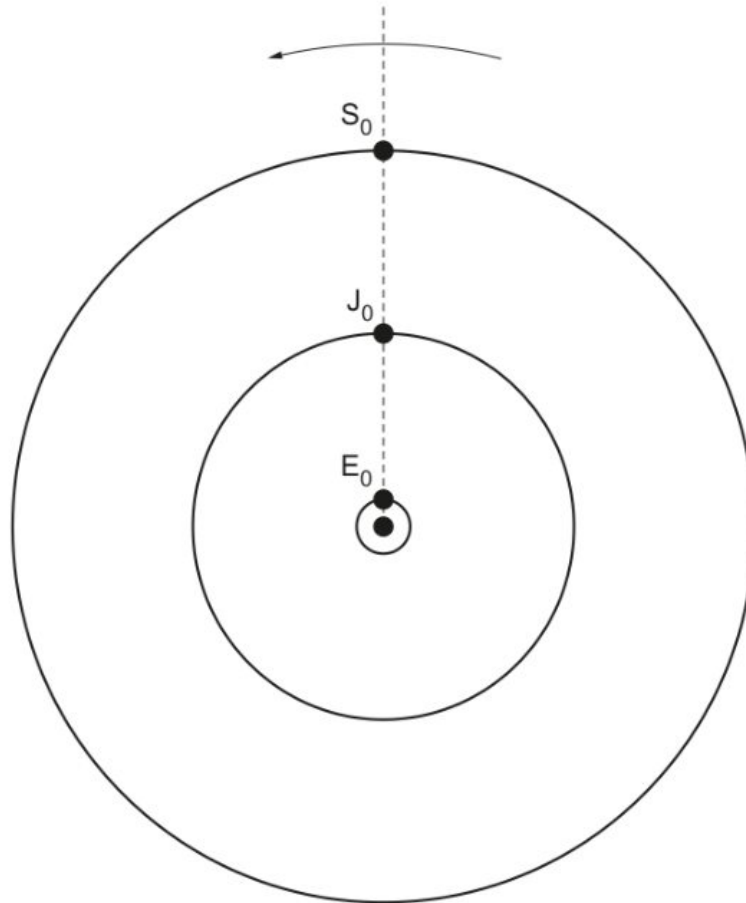


Fig. 9.1 (not to scale)

Assume that Saturn takes 30 years to orbit the Sun and that Jupiter takes 12 years to orbit the Sun.

- (a) On Fig. 9.1, mark the positions of Saturn and Jupiter 5.0 years after the original positions shown.

Label these positions S_1 and J_1 . Show your working.

- (b) (i) On Fig. 9.1, mark the positions of Saturn and Jupiter 20 years after the original positions shown in Fig. 9.1.

Label these positions S_2 and J_2 .

[1]

- (ii) State what is observed from the Earth's orbit after 20 years.

.....
 [1]

- (c) (i) Choose **two** words from the list to describe **each** planet.

gaseous large rocky small

Jupiter

Earth [1]

- (ii) The average density of Jupiter is much less than that of the Earth.
 The gravitational field strength at the surface of Jupiter is greater than that at the surface of the Earth.

Explain how these differences in density and in gravitational field strength are consistent with your answers to (c)(i).

density

.....

gravitational field strength

.....

[3]

- (d) The average density of Jupiter is 1300 kg/m^3 and its volume is $1.4 \times 10^{15} \text{ km}^3$.

Calculate the mass of Jupiter.

mass = [3]

[Total: 12]

- 9 Table 9.1 gives information about three planets in the Solar System.

Table 9.1

planet	mass / 10^{24} kg	average distance from Sun / 10^6 km	orbital period / days	gravitational field strength at surface
				N/kg
Earth	5.97	149.6	365.2	9.8
Jupiter	1898	778.6	4331	23.1
X	4.87	108.2	224.7	8.9

- (a) State the name of planet X.

..... [1]

- (b) Describe the relationship shown in Table 9.1 between the mass of a planet and the gravitational field strength at its surface.

.....
 [1]

- (c) Explain why 'distance from Sun' in Table 9.1 is an average value.

.....
 [1]

- (d) Show that the average orbital speed of the Earth is approximately 30 km/s.

[3]

[Total: 6]

- iv (a) The Solar System includes the Sun and planets.

State **two** other types of natural object that orbit the Sun.

1

2 [2]

- (b) State the shape of the orbits of the planets.

..... [1]

- (c) Fig. 10.1 shows the orbit of an object around the Sun. At point A, the object is closest to the Sun. At point B, the object is furthest away from the Sun.

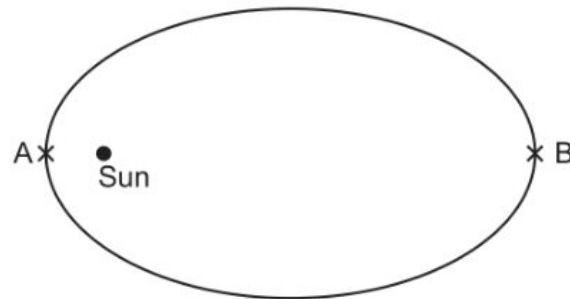


Fig. 10.1

State and explain the energy transfer as the object travels from point A to point B.

statement

.....

explanation

..... [2]

- (d) Jupiter is 7.8×10^{11} m from the Sun. The speed of light in a vacuum is 3.0×10^8 m/s.

Calculate the time taken for light from the Sun to reach Jupiter.

time = [2]

9 The Milky Way is the galaxy in which the Solar System is located.

(a) State what a galaxy is.

.....
 [1]

(b) The Milky Way has a diameter that is approximately equal to 100 000 light-years.

Determine this distance in kilometres (km).

distance = km [2]

(c) Astronomers determine the speed and distance from the Earth of a far galaxy that is moving away from the Earth.

(i) State **one** observation that allows the speed at which a galaxy is moving away to be determined.

.....
 [1]

(ii) State **one** different observation that is used to determine the distance to a far galaxy.

.....
 [1]

(iii) State how the speeds of galaxies and their distances from the Earth are related.

.....
 [1]

(iv) The best estimate for the Hubble constant H_0 is 2.2×10^{-18} per second.

Use this value to calculate an estimate for the age of the Universe.

age of the Universe = s [2]

[Total: 8]

- 11 (a) (i) State the name of **one** planet that has an orbit further away from the Sun than Venus.

..... [1]

- (ii) State the name of **one** planet that has an orbit closer to the Sun than Venus.

..... [1]

- (b) Venus has an average radius of orbit of 1.1×10^{11} m and an orbital period of 220 Earth days. Calculate the average orbital speed of Venus. Give your answer in m/s.

average orbital speed = m/s [3]

- (c) State the relationship between the orbital speeds of the planets and their distances from the Sun.

.....
..... [1]

- (d) Comets are balls of ice and dust. Some comets orbit the Sun.

State how the speed of a comet changes as it orbits the Sun.
Explain your answer using ideas about the conservation of energy.

You may include a labelled diagram in your answer.

.....
.....
..... [3]

[Total: 9]

- 10 Fig. 10.1 shows the path of the Earth as it orbits the Sun. X is a position on the Earth where scientists observe the apparent motion of the Sun throughout the year.

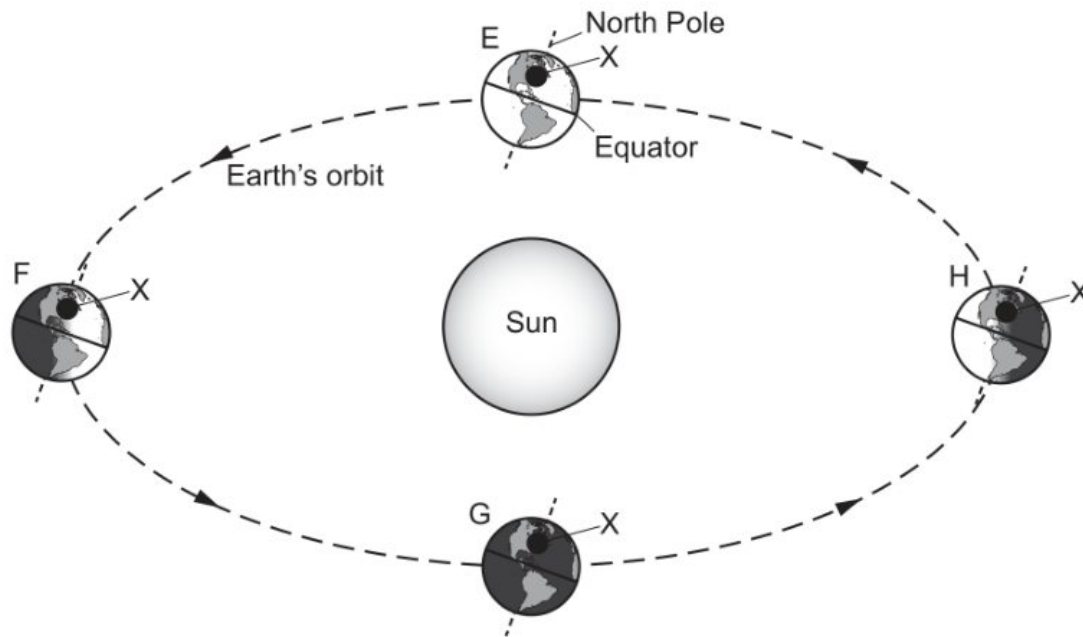


Fig. 10.1

- (a) Determine how many days it takes the Earth to move around its orbit from F to G. Explain your answer.

number of days =

explanation

[2]

- (b) Fig. 10.1 shows four positions E, F, G and H of the Earth in its orbit of the Sun.

(i) Identify the position of the Earth when it is summer at X. [1]

(ii) Identify the position of the Earth when it is winter at X. [1]

- (c) The orbital speed of the Earth around the Sun is approximately $3.0 \times 10^4 \text{ m/s}$.

Calculate the average radius of the Earth's orbit.

radius = [3]

- (d) Earth is a planet in the Solar System. State **one other** type of naturally occurring object that is present in the Solar System.

..... [1]

[Total: 8]

10 Jupiter and the Earth are planets in our Solar System.

(a) Describe the composition of Jupiter and the Earth.

Jupiter
the Earth [2]

(b) The gravitational field strength at the surface of the Earth is approximately 9.8 N/kg . The gravitational field strength at the surface of Jupiter is approximately 23 N/kg .

(i) Define gravitational field strength.

.....
..... [2]

(ii) State **one** factor which causes the difference between the gravitational field strength at the surface of Jupiter and the gravitational field strength at the surface of the Earth.

.....
..... [1]

(c) State and explain the difference between the orbital speed of Jupiter and the orbital speed of the Earth.

statement
explanation
..... [3]

[Total: 8]

- 10 (a) Table 10.1 shows data for planets A, B, C and D.

Table 10.1

planet	surface temperature /K	diameter /km	gravitational field strength N/kg
A	153	143 000	23.0
B	623	4 800	3.7
C	53	50 000	11.0
D	93	120 000	9.0

- (i) State and explain which one of these planets is closest to the Sun.

.....
 [2]

- (ii) Calculate the surface temperature of planet D in °C.

surface temperature = °C [1]

- (iii) An object falls through a height of 2 m on each of the planets in Table 10.1.

State and explain on which planet the object falls 2 m in the shortest time. Ignore any effect due to the atmosphere.

.....
 [2]

(b) Fig. 10.1 shows some of the stages in the life cycle of a massive star.

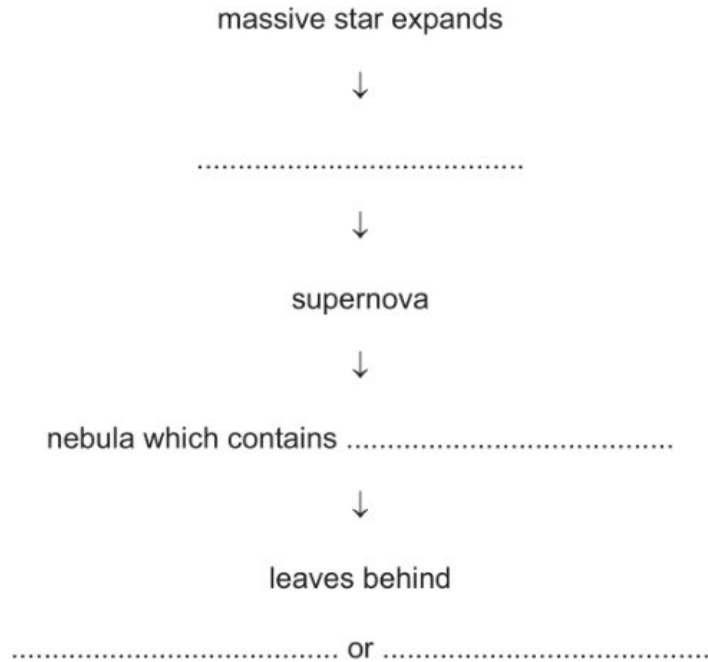


Fig. 10.1

- (i) Complete Fig. 10.1 by adding the correct terms. [4]
- (ii) State the quantity that the brightness of a supernova in a galaxy can be used to determine.

..... [1]

[Total: 10]

- 10 (a) Fill in the gaps in the description of how an accretion model explains the formation of the Solar System.

The planets nearest the Sun are small and

The planets furthest from the Sun are large and

All the planets were formed when a cloud of gas and dust collapsed due to

.....

The rotation of material in the cloud forms an

[4]

- (b) The time taken for light to travel from the Moon to the Earth is 1.3s.

Calculate the distance of the Moon from the Earth. Give your answer in m.

distance = m [3]

- (c) The Whirlpool galaxy is 2.3×10^7 light-years from the Earth. The current value of the Hubble constant is 2.2×10^{-18} per second.

Calculate the speed at which the Whirlpool galaxy is moving away from the Earth.

speed = [3]

[Total: 10]

- 9 (a) Cosmic microwave background radiation (CMBR) can be observed in every direction in space around us.

- (i) State when CMBR was produced.

..... [1]

- (ii) State and explain how the wavelength of this radiation has changed since it was formed.

.....

.....

..... [2]

- (b) Table 9.1 shows data about planets in our solar system.

Table 9.1

planet	average radius of orbit around the Sun / 10^6 km	time for one orbit / Earth years	average surface temperature / $^{\circ}\text{C}$
Venus	108	0.62	460
Earth	150	1.00	15
Mars	228	1.90	-65
Jupiter	778	12.00	-110
Saturn	1430	29.00	-140

- (i) The orbit of Venus is approximately circular.

Calculate the average orbital speed v of Venus. Give your answer in km/h.

orbital speed = km/h [3]

- (ii) State how the average surface temperature of a planet changes as its distance from the Sun increases.

..... [1]

[Total: 7]

6.5 Stars and the Universe

#	SOURCE	MARKS	DONE	MARK	REVIEW
01	Feb/March 2024 Paper 42 Question 11	-	☐	☐	☐

- 11 (a) State the condition required for a protostar to become a stable star.

.....
.....
..... [1]

- (b) (i) Define the Hubble constant.

.....
.....
..... [2]

- (ii) The current estimate for the Hubble constant is 2.2×10^{-18} per second.

State the equation which gives an estimate for the age of the Universe.

..... [1]

- (iii) Calculate an estimate for the age of the Universe.

estimate of age = s [1]

[Total: 5]

6.6 Stars, galaxies and the Universe

#	SOURCE	MARKS	DONE	MARK	REVIEW
01	May/June 2023 Paper 43 Question 10	-	☐	☐	☐
02	Oct/Nov 2023 Paper 41 Question 10	-	☐	☐	☐
03	Oct/Nov 2023 Paper 42 Question 10	-	☐	☐	☐
04	Oct/Nov 2023 Paper 43 Question 10	-	☐	☐	☐
05	May/June 2024 Paper 41 Question 9	-	☐	☐	☐
06	May/June 2024 Paper 42 Question 11	-	☐	☐	☐
07	May/June 2024 Paper 43 Question 11	-	☐	☐	☐
08	Oct/Nov 2024 Paper 43 Question 10	-	☐	☐	☐
09	May/June 2025 Paper 43 Question 10	-	☐	☐	☐

10 (a) Show that 1 light-year = 9.5×10^{15} m.

[4]

(b) (i) State **one** measurement that is taken when determining the speed v at which a galaxy is moving away from the Earth.

.....
..... [1]

(ii) Write down an equation relating v and the distance d of a far galaxy.

..... [1]

(iii) State how the distance d of a far galaxy can be determined other than by using the equation in (ii).

.....
..... [1]

[Total: 7]

10 The Milky Way is one of many billions of galaxies. Each galaxy contains many billions of stable stars.

- (a) Stable stars transfer energy into space by emitting electromagnetic radiation from their surfaces.

Describe what happens in the core of a stable star to release energy that is eventually transferred into space.

.....

 [3]

- (b) On the Earth, light from a distant galaxy is observed and analysed by astronomers. This information is used to determine the speed at which the galaxy is moving away from the Earth.

- (i) Describe how the observed light is different from when it was emitted.

.....
 [2]

- (ii) State the quantity that astronomers use to determine the speed at which the galaxy is moving away.

..... [1]

- (c) The Hubble constant H_0 is equal to 2.2×10^{-18} per second.

- (i) Calculate the distance from the Earth of a galaxy that is moving away at a speed of 1.3×10^7 m/s.

distance = [2]

- (ii) Calculate an estimate for the age of the Universe. Give your answer in years.

age of the Universe = years [2]

[Total: 10]

10 Complete the sentences about the life cycle of stars.

(a) Protostars are formed from
..... [1]

(b) A protostar becomes a stable star when
.....
is balanced by
..... [2]

(c) The initial fuel used to power nuclear reactions in stars is [1]

(d) Stars that are approximately the same size as the Sun become red giant stars which then
form a
with a white dwarf star at its centre. [1]

[Total: 5]

- 10 (a) (i) 1. State what is represented in space physics by the symbol H_0 .
- [1]
2. Write down the equation that defines H_0 in terms of the speed that a far galaxy is moving away from the Earth and its distance from the Earth.
- [1]
- (ii) The numerical value of H_0 is 2.2×10^{-18} . State the unit of H_0 .
- [1]
- (iii) Use this value of H_0 to determine an estimate for the age of the Universe in seconds.

age of the Universe = s [2]

- (b) State when cosmic microwave background radiation (CMBR) was formed and where we detect it coming from.

.....

.....

.....

..... [2]

[Total: 7]

- 9 The Sun is one of many billions of stars in the Milky Way. The Sun emits a very large quantity of energy as electromagnetic radiation.

- (a) State the **three** regions of the electromagnetic spectrum in which the Sun emits the most energy.

1

2

3 [2]

- (b) Electromagnetic radiation from the Sun travels at a speed of $3.0 \times 10^8 \text{ m/s}$. The radiation takes 500 s to reach the Earth.

Calculate the distance from the Sun to the Earth.

distance = [2]

- (c) Approximately 4.6 billion years ago, the Sun formed from an interstellar cloud of gas and became a stable star.

- (i) Describe and explain what happens as an interstellar cloud of gas forms a protostar.

.....

.....

..... [2]

- (ii) Describe and explain what happens as a protostar becomes a stable star.

.....

.....

.....

..... [3]

[Total: 9]



11 (a) Name the galaxy that contains the Sun.

..... [1]

(b) Light observed from distant galaxies is redshifted.

State the theory of the Universe that this observation supports.

..... [1]

(c) Cosmic microwave background radiation (CMBR) is observed at all points in space.

(i) State when this radiation was produced.

..... [1]

(ii) Explain why this radiation is now in the microwave region of the electromagnetic spectrum.

.....
..... [2]

[Total: 5]

11 A galaxy is approximately 1.2×10^{26} m from the Earth.

(a) Scientists observe light from the distant galaxy.

The wavelength of the observed light is longer than the wavelength of the light emitted from the galaxy.

State the name of this effect.

..... [1]

(b) (i) State the current estimate for the Hubble constant H_0 .

$H_0 =$ [1]

(ii) Calculate the speed at which the galaxy is moving away from the Earth.

speed = [2]

(iii) Scientists have measured the speeds at which distant galaxies are moving away from the Earth and their distances from the Earth.

These measurements suggest that all the Universe was once at a single point.

Explain why.

.....
.....
..... [2]

[Total: 6]

- 10 (a)** Stars more massive than the Sun can eventually form black holes.

Describe how a black hole can be formed from a more massive star.

.....

.....

.....

.....

..... [3]

- (b)** The star system V404 Cygni contains a black hole. The system is approximately 7800 light-years from the Earth.

- (i)** Describe what is meant by a light-year.

.....

..... [1]

- (ii)** Calculate the approximate distance from V404 Cygni to the Earth in km.

distance = km [2]

[Total: 6]

(a) Define a light year in words.

..... [1]

(b) It takes light 490 s to travel from the Sun to the Earth.

Calculate the distance from the Sun to the Earth.

distance = [2]

(c) Fig. 10.1 is a scatter graph showing how the speed of galaxies moving away from the Earth varies with their distances from the Earth.

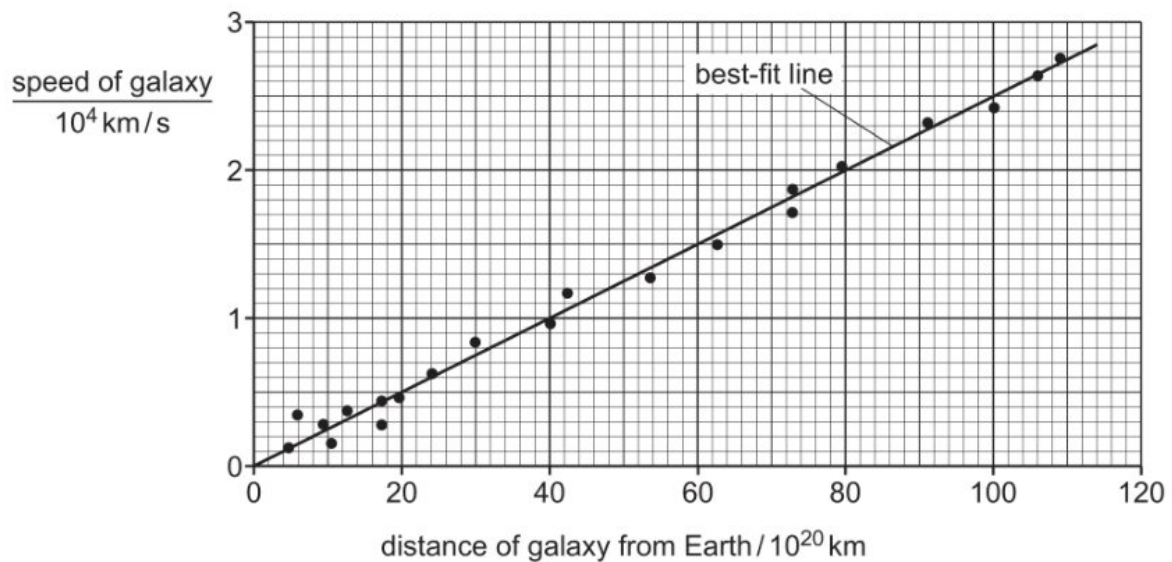


Fig. 10.1

A scientist draws a best-fit line on the scatter graph.

Use this best-fit line to determine a value for the Hubble constant.

Hubble constant = [3]

(d) The speed of a receding galaxy can be estimated using redshift.

Describe what is meant by redshift.

.....

.....

.....

..... [2]

[Total: 8]

6.7 Space physics - mixed

#	SOURCE	MARKS	DONE	MARK	REVIEW
01	Oct/Nov 2025 Paper 41 Question 10	-	☐	☐	☐

10 Fig. 10.1 shows the orbit of the Earth and the orbit of a comet around the Sun.

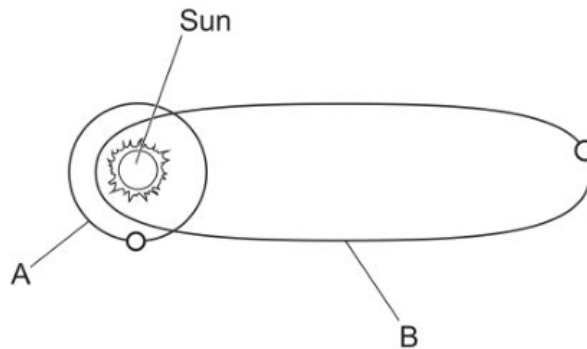


Fig. 10.1

- (a) State which orbit, A or B, is the orbit of the comet.
Explain your answer.

orbit of comet is

explanation
..... [1]

- (b) Describe and explain how the motion of the comet changes as it orbits the Sun.

.....
.....
..... [2]

- (c) At one position in its orbit, the comet is 6.6×10^{-6} light-years away from the Earth.

- (i) State the meaning of light-year.

.....
..... [1]

- (ii) Determine the distance in metres between the comet and the Earth.

distance = m [2]

[Total: 6]